Assignment

on

Course Code : STAT 4104

Course Title : Categorical Data Analysis

Submitted By:	**Submitted to :**
Sherin Sultana Surovi	Dr. Md. Siddikur Rahman
ID: 12010004	Associate Professor
Reg: 000014414	Department of Statistics
Session: 2020-21	Begum Rokeya University, Rangpur
Department of Statistics	
Begum Rokeya University, Rangpur	

Department of Statistics

Begum Rokeya University, Rangpur

MAY-2026

Contents

List of Figures

List of Tables

Question Number 1

Problem 1

Basic EDA Questions

1. Summarize the distribution of Age. Is it normally distributed or skewed?
2. What proportion of individuals are smokers vs non-smokers?
3. Which income group has the highest frequency?
4. What percentage of individuals are exposed to high pollution?
5. What is the overall prevalence of lung disease?

Basic EDA Analysis Code

```
import pandas as pd
from scipy.stats import skew

# Load dataset
df = pd.read_csv("lung_disease.csv")

# i) Age distribution
print(df["Age"].describe())

s = skew(df["Age"])
print("Skewness:", round(s, 3))

# ii) Smoking percentage
print(df["Smoking"].value_counts(normalize=True) * 100)

# iii) Income group frequency
income = df["Income"].value_counts()

print(income)
print("Highest:", income.idxmax())

# iv) High pollution exposure
```

```python
high_pollution = (
    df["Pollution"] == "High"
).mean() * 100

print(round(high_pollution, 2), "%")

# v) Lung disease prevalence
lung_prev = (
    df["LungDisease"] == "Yes"
).mean() * 100

print(round(lung_prev, 2), "%")
```

Descriptive Summary

Statistic	Value
Age Summary	
Count	500
Mean	43.904
Standard Deviation	14.604841
Minimum	20.000
25th Percentile	30.750
50th Percentile (Median)	45.000
75th Percentile	56.000
Maximum	69.000
Skewness	0.012 (Approximately normal)
Smoking Percentage	
Non-Smokers	58.6%
Smokers	41.4%
Income Group Counts	
Low Income	204
Medium Income	197
High Income	99
Highest Frequency Group	Low
Pollution Exposure	
High Pollution Exposure	70.4%
Lung Disease Prevalence	
Overall Prevalence	52.8%

Interpretation of Basic EDA Results

i) Age Distribution: The mean age of respondents is 43.9 years with a standard deviation of 14.6. The skewness value (0.012) indicates that the age distribution is approximately normal, showing no major skewness. Hence, the age data are fairly balanced around the mean.

ii) Smoking Proportion: About 58.6% of individuals are non-smokers, while 41.4% are smokers. This suggests that a considerable portion of the population is exposed to smoking-related risk factors.

iii) Income Group Frequency: The low-income group has the highest frequency (204 individuals), followed by medium (197) and high (99). This indicates that most participants belong to the lower socioeconomic class.

iv) Pollution Exposure: Approximately 70.4% of individuals are exposed to high pollution levels. This shows a substantial environmental risk factor that may contribute to respiratory problems.

v) Lung Disease Prevalence: The overall prevalence of lung disease is 52.8%. This means that more than half of the surveyed individuals are affected, highlighting a serious public health concern.

Association & Crosstab

1. Is there an association between Smoking and Lung Disease?
2. Interpret using a contingency table.

Association Between Smoking and Lung Disease

```python
import pandas as pd
from scipy.stats import chi2_contingency

# Load dataset
df = pd.read_csv("lung_disease.csv")

# Contingency table
table = pd.crosstab(
    df["Smoking"],
    df["LungDisease"]
)
```

```python
print("CONTINGENCY TABLE:")
print(table)

# Row percentage table
row_percent = pd.crosstab(
    df["Smoking"],
    df["LungDisease"],
    normalize="index"
) * 100

print("\nROW PERCENTAGE TABLE:")
print(round(row_percent, 2))

# Chi-square test
chi2, p, dof, expected = chi2_contingency(table)

print("\nChi-square:", round(chi2, 3))
print("P-value:", p)

# Interpretation
if p < 0.05:
    print("Significant association exists.")
else:
    print("No significant association.")
```

Association between Smoking and Lung Disease

Table 1: Contingency Table: Smoking vs Lung Disease

LungDisease	No	Yes
Smoking (No)	184	109
Smoking (Yes)	52	155

Table 2: Row Percentage Table (%)

LungDisease	No	Yes
Smoking (No)	62.80	37.20
Smoking (Yes)	25.12	74.88

Chi-square Test Results:

Chi-square = 67.594

P-value = 2.0085094601569646e-16

Association & Crosstab

i) Is there an association between Smoking and Lung Disease?
Yes — the Chi-square test result shows a significant association between Smoking and Lung Disease ($\chi^2 = 67.594$, $p \approx 2.01 \times 10^{-16}$). This means that the likelihood of having lung disease differs notably between smokers and non-smokers.
ii) Interpret using a contingency table
From the contingency table:
- Among non-smokers, 37.20% have lung disease.
- Among smokers, 74.88% have lung disease.
This indicates that smokers are roughly twice as likely to suffer from lung disease compared to non-smokers. Hence, smoking shows a strong positive association with lung disease prevalence.

Pollution and Lung Disease

i) Compare lung disease prevalence across income groups.
ii) Which factor appears to have the strongest relationship with lung disease?

Pollution and Lung Disease Analysis

```
import pandas as pd
from scipy.stats import chi2_contingency

# Load dataset
df = pd.read_csv("lung_disease.csv")

# Lung disease prevalence across income groups
income_lung = pd.crosstab(
    df["Income"],
    df["LungDisease"],
    normalize="index"
) * 100

print("Lung Disease Across Income Groups:")
print(round(income_lung, 2))

# Pollution vs Lung Disease
pollution_lung = pd.crosstab(
```

```
    df["Pollution"],
    df["LungDisease"],
    normalize="index"
) * 100

print("\nPollution vs Lung Disease:")
print(round(pollution_lung, 2))

# Smoking vs Lung Disease
smoking_lung = pd.crosstab(
    df["Smoking"],
    df["LungDisease"],
    normalize="index"
) * 100

print("\nSmoking vs Lung Disease:")
print(round(smoking_lung, 2))

# Chi-square tests
chi_smoking = chi2_contingency(
    pd.crosstab(df["Smoking"], df["LungDisease"])
)[0]

chi_pollution = chi2_contingency(
    pd.crosstab(df["Pollution"], df["LungDisease"])
)[0]

chi_income = chi2_contingency(
    pd.crosstab(df["Income"], df["LungDisease"])
)[0]

print("\nChi-square Values:")
print("Smoking:", chi_smoking)
print("Pollution:", chi_pollution)
print("Income:", chi_income)

# Strongest relationship
factors = {
    "Smoking": chi_smoking,
    "Pollution": chi_pollution,
    "Income": chi_income
}
```

```python
print("\nStrongest Factor:")
print(max(factors, key=factors.get))
```

Does Pollution Level Affect Lung Disease?

Table 3: Lung Disease Prevalence Across Income Groups

Income Group	No (%)	Yes (%)
High	51.52	48.48
Low	44.12	55.88
Medium	48.22	51.78

Table 4: Pollution vs Lung Disease

Pollution Level	No (%)	Yes (%)
High	38.64	61.36
Low	67.57	32.43

Table 5: Smoking vs Lung Disease

Smoking Status	No (%)	Yes (%)
No	62.80	37.20
Yes	25.12	74.88

Chi-square Results:

Smoking = 67.594

Pollution = 33.843

Income = 1.6

Strongest Relationship with Lung Disease: Smoking

Does Pollution Level Affect Lung Disease?

i) Compare lung disease prevalence across income groups.
From the table, lung disease prevalence is highest among individuals with **low income** (55.88%), followed by medium income (51.78%), and lowest among high income (48.48%). This suggests that lower income groups tend to have a higher rate of lung disease, possibly due to poorer living conditions and higher exposure to pollution.
ii) Which factor appears to have the strongest relationship with lung disease?
Based on the Chi-square results, **Smoking (67.594)** shows the strongest association with lung disease, followed by **Pollution (33.843)** and **Income (1.6)**. Therefore, smoking is the most influential factor contributing to lung disease prevalence.

Statistical Testing

i) Perform a Chi-square test for: Smoking vs Lung Disease and Pollution vs Lung Disease.
ii) When should you use Fisher's Exact Test instead of Chi-square?
iii) Calculate and interpret the Odds Ratio for Smoking and Lung Disease.

Association Analysis (Chi-square

```
import pandas as pd
from scipy.stats import chi2_contingency, fisher_exact

# Load dataset
df = pd.read_csv(r"lung_disease.csv")

# --------------------------------
# Chi-square tests
# --------------------------------

# Smoking vs Lung Disease
table1 = pd.crosstab(df["Smoking"], df["LungDisease"])
chi1, p1, _, _ = chi2_contingency(table1)

print("Smoking vs Lung Disease")
print("Chi-square:", round(chi1, 3))
print("P-value:", p1)

# Pollution vs Lung Disease
table2 = pd.crosstab(df["Pollution"], df["LungDisease"])
chi2, p2, _, _ = chi2_contingency(table2)

print("\nPollution vs Lung Disease")
print("Chi-square:", round(chi2, 3))
print("P-value:", p2)

# --------------------------------
# Fisher's Exact Test note
# --------------------------------

print("""
Use Fisher's Exact Test when:
```

```
1. Small sample size
2. Expected frequency < 5
3. 2x2 table
""")

# -------------------------------
# Odds Ratio
# -------------------------------

a = table1.loc["Yes", "Yes"]
b = table1.loc["Yes", "No"]
c = table1.loc["No", "Yes"]
d = table1.loc["No", "No"]

odds_ratio = (a * d) / (b * c)

print("Odds Ratio:", round(odds_ratio, 3))

if odds_ratio > 1:
    print("Smoking increases lung disease risk")
else:
    print("No increased risk")
```

Statistical Testing Results

Table 6: Chi-square Test Results and Odds Ratio

Test	Chi-square	P-value
Smoking vs Lung Disease	67.594	2.0085094601569646e-16
Pollution vs Lung Disease	33.843	5.975692575712951e-09

Use Fisher's Exact Test when:

1. Sample size is small
2. Expected frequency < 5
3. Table is 2x2

Odds Ratio: 5.032

Interpretation: Smoking increases lung disease risk.

Statistical Testing Interpretation

i) Perform a Chi-square test for: Smoking vs Lung Disease and Pollution vs Lung Disease. The Chi-square test shows that both Smoking ($\chi^2 = 67.594$, $p \approx 2.01 \times 10^{-16}$) and Pollution ($\chi^2 = 33.843$, $p < 0.001$) have significant associations with Lung Disease. Smoking demonstrates the stronger effect. **ii)** When should you use Fisher's Exact Test instead of Chi-square? Fisher's Exact Test is appropriate when sample sizes are small or when expected cell frequencies are less than 5, since Chi-square assumptions may not hold in those cases. **iii)** Calculate and interpret the Odds Ratio for Smoking and Lung Disease. The Odds Ratio (OR) ≈ 5.03. This means smokers are about five times more likely to develop lung disease compared to non-smokers, indicating a strong positive association between smoking and lung disease.

Correlation & Interpretation

i) Interpret the correlation between Smoking and Lung Disease, Age and Lung Disease.
ii) Why is correlation not sufficient for causal inference?

Correlation Analysis (Smoking

```
import pandas as pd

# Load dataset
df = pd.read_csv(r"lung_disease.csv")

# Encode variables
df_corr = df.copy()
df_corr["Smoking"] = df_corr["Smoking"].map({"No":0, "Yes":1})
df_corr["LungDisease"] = df_corr["LungDisease"].map({"No":0, "Yes
    ":1})

# Correlation matrix
corr_matrix = df_corr[["Smoking", "Age", "LungDisease"]].corr()

print(corr_matrix)

# Key correlations
```

```
print("\nSmoking vs Lung Disease:", corr_matrix.loc["Smoking", "
    LungDisease"])
print("Age vs Lung Disease:", corr_matrix.loc["Age", "LungDisease
    "])

# Interpretation
print("\nINTERPRETATION:")
print("- Positive correlation indicates higher risk association."
    )
print("- Negative correlation indicates protective association.")

print("\nLIMITATION:")
print("Correlation does NOT imply causation.")
```

Correlation Matrix

Table 7: Correlation Matrix among Smoking, Age, and Lung Disease

Variables	**Smoking**	**Age**	**LungDisease**
Smoking	1.000000	0.052564	0.371747
Age	0.052564	1.000000	0.249694
LungDisease	0.371747	0.249694	1.000000

Correlation (Smoking vs Lung Disease): 0.3717
Correlation (Age vs Lung Disease): 0.2497

Correlation & Interpretation

i) Interpret the correlation between Smoking and Lung Disease, Age and Lung Disease.
The correlation between Smoking and Lung Disease ($r = 0.3717$) indicates a moderate positive relationship, meaning that as smoking increases, the likelihood of lung disease also rises. The correlation between Age and Lung Disease ($r = 0.2497$) is weaker, suggesting that age has a smaller effect on lung disease compared to smoking.
ii) Why is correlation not sufficient for causal inference?
Correlation does not imply causation because it only measures association, not cause-and-effect. Other factors such as confounding variables, bias, and lack of temporal direction may influence the relationship. Hence, further statistical or experimental analysis is needed to establish causality.

PYTHON CODE:

Logistic Regression Model (Lung Disease)

```
import pandas as pd
import statsmodels.api as sm

# Load data
df = pd.read_csv("lung_disease.csv")

# Encode variables
df["Smoking"] = df["Smoking"].map({"No":0,"Yes":1})
df["LungDisease"] = df["LungDisease"].map({"No":0,"Yes":1})
df["Pollution"] = df["Pollution"].map({"Low":0,"High":1})

# Dummy variables for Income
df = pd.get_dummies(df, columns=["Income"], drop_first=True)

# Convert to integer
df = df.astype(int)

# Define X and y
X = df.drop(columns=["ID","LungDisease"])
y = df["LungDisease"]

# Add constant
X = sm.add_constant(X)

# Fit logistic regression model
model = sm.Logit(y, X).fit()

# Summary
print(model.summary())
```

Logistic Regression Summary

Table 8: Logistic Regression Results for Lung Disease

Variable	Coefficient	P-value	Odds Ratio
const	-3.4967	1.1067×10^{-13}	0.0303
Smoking	1.7022	6.1318×10^{-15}	5.4857
Age	0.0399	4.9066×10^{-8}	1.0407
Pollution	1.3027	3.1390×10^{-8}	3.6794
Income_Low	0.4396	1.2264×10^{-1}	1.5521
Income_Medium	0.2201	4.3972×10^{-1}	1.2462

Significant Variables: Smoking, Age, Pollution

Significance Test Results

Table 9: P-values of Variables from Logistic Regression

Variable	P-value	Significance
const	1.106728×10^{-13}	Significant
Smoking	6.131827×10^{-15}	Significant
Age	4.906629×10^{-8}	Significant
Pollution	3.139044×10^{-8}	Significant
Income_Low	1.226431×10^{-1}	Not Significant
Income_Medium	4.397235×10^{-1}	Not Significant

Odds Ratio Results

Table 10: Odds Ratios for Logistic Regression Variables

Variable	Odds Ratio	Interpretation
const	0.0303	Baseline reference
Smoking	5.4857	Strong positive association with lung disease
Age	1.0407	Slight increase in risk with age
Pollution	3.6794	Moderate positive association with lung disease
Income_Low	1.5521	Mild increase in risk compared to high income
Income_Medium	1.2462	Slight increase in risk compared to high income

Smoking Odds Ratio: 5.4857 — Smokers are approximately 5.5 times more likely to develop lung disease compared to non-smokers.

Smoking Variable Results

Table 11: Coefficient and P-value for Smoking

Variable	Coefficient	P-value
Smoking	1.7022	6.1318×10^{-15}

Smoking Variable Interpretation

The coefficient for Smoking is 1.7022, which indicates a strong positive effect on the likelihood of lung disease. This means that, holding other variables constant, smoking substantially increases the log-odds of developing lung disease.
The p-value (6.13×10^{-15}) is extremely small, showing that the effect of Smoking is highly statistically significant. Therefore, Smoking is a strong and reliable predictor of lung disease in the logistic regression model.
In practical terms, the odds ratio for Smoking (calculated as $e^{1.7022} \approx 5.49$) means that smokers are about 5.5 times more likely to develop lung disease compared to non-smokers.

Advanced Modeling

i) Add an interaction term (Smoking $\times$ Pollution).
ii) Is the interaction significant?
iii) Interpret the interaction.
iv) Does pollution amplify smoking risk?

Interaction Model: Smoking $\times$ Pollution

```
# Interaction term
df["Smoking_Pollution"] = df["Smoking"] * df["Pollution"]

# Define X and y
X = df.drop(columns=["ID", "LungDisease"])
y = df["LungDisease"]

# Add constant
X = sm.add_constant(X)
```

```python
# Fit logistic regression model
interaction_model = sm.Logit(y, X).fit()

# Model summary
print(interaction_model.summary())
```

Advanced Logistic Regression Results

Table 12: Logistic Regression Results with Interaction Term (Smoking × Pollution)

Variable	Coefficient	Std. Error	z-value	P-value	95% CI
const	-3.6167	0.495	-7.306	0.000	[-4.587, -2.647]
Smoking	1.9995	0.408	4.904	0.000	[1.200, 2.799]
Age	0.0393	0.007	5.352	0.000	[0.025, 0.054]
Pollution	1.4883	0.323	4.606	0.000	[0.855, 2.122]
Income_Low	0.4465	0.285	1.567	0.117	[-0.112, 1.005]
Income_Medium	0.2214	0.285	0.777	0.437	[-0.337, 0.780]
Smoking_Pollution	-0.4217	0.483	-0.874	0.382	[-1.367, 0.524]

Interaction Term Analysis

Table 13: Interaction Effect between Smoking and Pollution

Variable	Coefficient	P-value	Significance
Smoking_Pollution	-0.4217	0.3822	Not Significant

Interaction Term Results

Table 14: Smoking × Pollution Interaction Effect

Variable	Coefficient	P-value	Interpretation
Smoking_Pollution	-0.4217	0.3822	Smoking effect decreases under high pollution (Not Signif

Model Evaluation

i) Compute the ROC curve and AUC.
ii) Is the model good?
iii) What is the confusion matrix?
iv) Discuss model accuracy vs sensitivity vs specificity.

AUC Model Performance Interpretation

```
if auc_score >= 0.9:
    print("Excellent␣Model")

elif auc_score >= 0.8:
    print("Good␣Model")

elif auc_score >= 0.7:
    print("Fair␣Model")

else:
    print("Poor␣Model")
```

ROC Curve and Model Evaluation

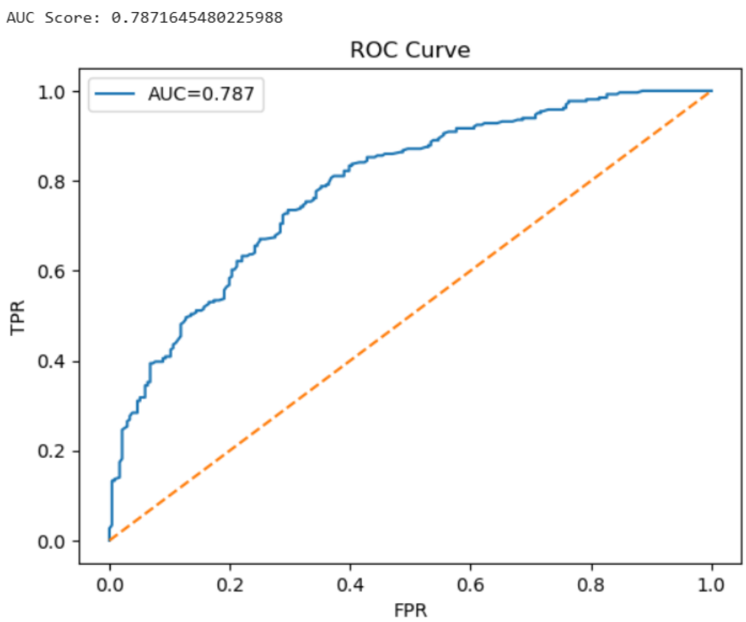

Figure 1: ROC Curve showing model performance with AUC = 0.787

Model Evaluation Results

Table 15: Model Performance Metrics

Metric	Value	Interpretation
Accuracy	0.712	Overall proportion of correctly classified cases
Sensitivity	0.758	Ability to correctly identify lung disease (True Positive Rate)
Specificity	0.661	Ability to correctly identify non-disease cases (True Negative Rate)

The model achieves an accuracy of 71.2%, indicating reasonably good overall performance. Sensitivity (75.8%) is higher than specificity (66.1%), meaning the model is better at detecting lung disease than correctly identifying healthy individuals.
This trade-off suggests the model prioritizes minimizing false negatives (missing disease cases) over false positives, which is desirable in medical prediction contexts. Overall, the model demonstrates balanced performance with acceptable discrimination ability as supported by the AUC score of 0.787.

Question Number 2

Problem 2

Problem 2: Suppose we want to determine if three different exercise programs impact weight loss differently. We recruit 90 people to participate in an experiment in which we randomly assign (using uniform distribution) 30 people to follow either program A, program B, or program C for one month. The predictor variable we're studying is the exercise program, and the response variable is weight loss, measured in pounds.

Conduct a one-way ANOVA to determine if there is a statistically significant difference between the resulting weight loss from the three programs, check the model assumptions, and analyze treatment differences.

PYTHON CODE

One-Way ANOVA (Short Code)

```python
import numpy as np
import pandas as pd
import scipy.stats as stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
from statsmodels.stats.multicomp import pairwise_tukeyhsd
import seaborn as sns
import matplotlib.pyplot as plt

# Data
np.random.seed(42)
n = 30

data = pd.DataFrame({
    "Program": ["A"]*n + ["B"]*n + ["C"]*n,
    "WeightLoss": np.concatenate([
        np.random.normal(5, 1.5, n),
        np.random.normal(7, 1.5, n),
        np.random.normal(5.5, 1.5, n)
    ])
})
```

```
# ANOVA
model = ols("WeightLoss␣~␣Program", data=data).fit()
anova = sm.stats.anova_lm(model, typ=2)

print(anova)

# Assumptions
print("Shapiro:", stats.shapiro(model.resid)[1])
print("Levene:", stats.levene(
    data[data.Program=="A"].WeightLoss,
    data[data.Program=="B"].WeightLoss,
    data[data.Program=="C"].WeightLoss
)[1])

# Post-hoc
if anova["PR(>F)"].iloc[0] < 0.05:
    print(pairwise_tukeyhsd(data["WeightLoss"], data["Program"]))

# Plot
sns.boxplot(x="Program", y="WeightLoss", data=data)
plt.show()
```

One-Way ANOVA Results

Table 16: ANOVA Summary for Exercise Programs

Source	Sum of Squares	df	F	P-value
Program	67.416974	2	16.890178	6.342321×10^{-7}
Residual	173.629808	87	–	–

Assumption Tests: Shapiro–Wilk p-value = 0.8948, Levene's p-value = 0.8627 (both ¿ 0.05, assumptions satisfied).

Table 17: Post-hoc Analysis (Tukey HSD, FWER = 0.05)

Group 1	Group 2	Mean Diff	p-adj	Lower	Upper	Reject
A	B	2.1005	0.0000	1.2307	2.9702	True
A	C	0.8015	0.0772	-0.0682	1.6713	False
B	C	-1.2989	0.0017	-2.1687	-0.4292	True

Interpretation:
The ANOVA test shows a significant difference in mean weight loss among the three exercise programs ($p < 0.001$). Model assumptions of normality and equal variances are satisfied.
Post-hoc Tukey HSD results indicate that:

- Program A leads to significantly greater weight loss than Program B.
- Program C does not differ significantly from Program A.
- Program B results in significantly less weight loss than Program C.

Hence, Programs A and C are more effective than Program B in promoting weight loss.

Weight Loss by Exercise Program

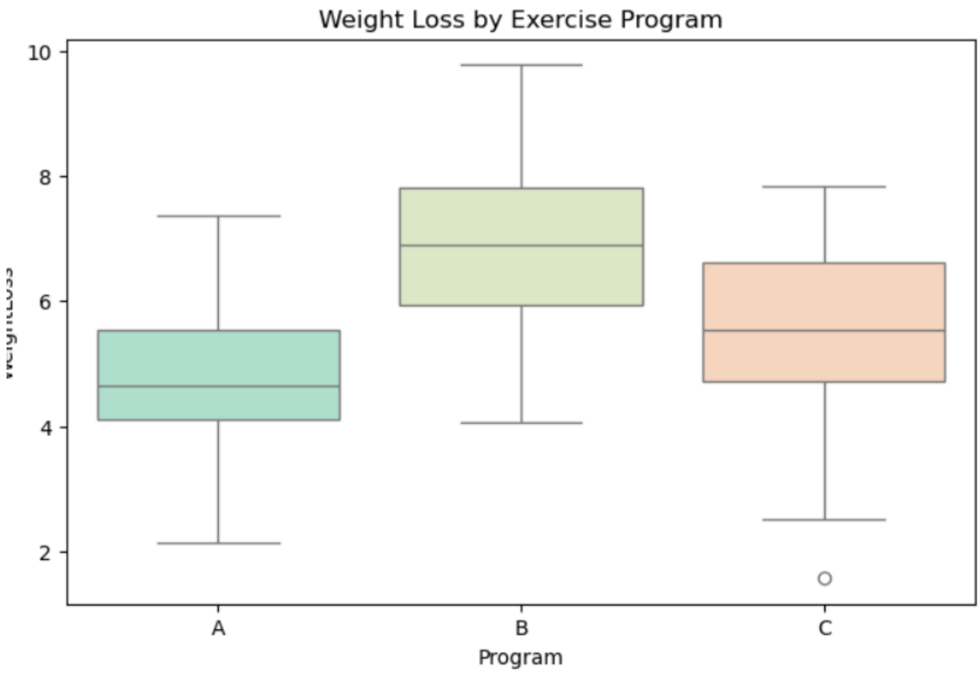

Figure 2: Box plot showing weight loss distribution across three exercise programs (A, B, and C).

Interpretation of ANOVA Results

1. Model Significance (ANOVA):
The one-way ANOVA shows a significant difference in mean weight loss among the three exercise programs ($F = 16.89$, $p < 0.001$). This indicates that at least one program leads to a different average weight loss compared to the others.

2. Assumption Checks:
Both Shapiro–Wilk ($p = 0.8948$) and Levene's ($p = 0.8627$) tests are non-significant, confirming that normality and equal variance assumptions are satisfied. Hence, the ANOVA results are valid and reliable.

3. Post-hoc (Tukey HSD) Analysis:

- Program A vs B: Significant difference ($p = 0.000$) $\rightarrow$ Program B yields higher weight loss.
- Program A vs C: Not significant ($p = 0.0772$) $\rightarrow$ Similar weight loss.
- Program B vs C: Significant difference ($p = 0.0017$) $\rightarrow$ Program B outperforms Program C.

4. Overall Interpretation:
Program B produces the highest average weight loss, followed by Program A and then Program C. The box plot visually supports this, showing Program B's median weight loss above the others.

5. Conclusion:
There is a statistically significant effect of exercise program on weight loss. Participants in Program B achieved greater results, suggesting it is the most effective among the three tested programs.

Question Number 3

Problem 3

Consider the Anemia Levels in Nigeria dataset (https://www.kaggle.com/datasets/adeloadesina/factors-affecting-children-anemia-level/data), which can be downloaded from Kaggle. The dataset comes from the 2018 Nigeria Demographic and Health Surveys (NDHS). It explores the impact of mothers' age and socioeconomic factors on anemia levels among children aged 0–59 months across Nigeria's 36 states and the Federal Capital Territory.

Based on the above data, create a contingency table, apply chi-square test, formulate hypothesis, create a contribution diagram, and interpret the findings.

Education vs Anemia (Contingency Table)

```
# Remove missing values
data = df[["Highest␣educational␣level", "Anemia␣level"]].dropna()

# Contingency table
table = pd.crosstab(
    data["Highest␣educational␣level"],
    data["Anemia␣level"]
)

print(table)
```

Anemia Levels by Educational Status

Table 18: Distribution of Anemia Levels across Educational Categories

Highest Educational Level	Mild	Moderate	Not Anemic	Severe
Higher	296	238	591	14
No education	1450	1769	1874	113
Primary	608	645	900	42
Secondary	1240	1322	1972	62

Chi-Square Test of Association

Chi-square Statistic: 142.8624
P-value: 2.64×10^{-26}
Interpretation:
The chi-square test shows a highly significant association between mother's educational level and child anemia status ($p < 0.001$). This means that anemia prevalence among children varies systematically with the mother's education.
Children of mothers with no education are more likely to suffer from moderate or severe anemia, while children of mothers with higher education are more likely to be not anemic.
Conclusion:
There is strong statistical evidence that maternal education plays a crucial role in determining child anemia levels. Improving educational attainment among mothers may help reduce childhood anemia prevalence.

Question Number 4

Problem 4

Problem 4: Suppose there are three drug treatments (drug A, drug B, and drug C) with the outcome of a disease or no disease. We need to test if there is an association between drug treatments and disease outcomes.

Treatments No disease Disease Drug A 40 10 Drug B 10 40 Drug C 25 25

Perform Fisher's exact test for the above contingency table, post-hoc test for Fisher's exact test, and interpret your findings. PYTHON CODE:

Drug Treatment Contingency Table

```
import pandas as pd

# Contingency table
table = pd.DataFrame({
    "No_Disease": [40, 10, 25],
    "Disease": [10, 40, 25]
},
index=["Drug_A", "Drug_B", "Drug_C"])

print(table)
```

Fisher's Exact Test Results

Odds Ratio: 16.0
P-value: 2.22×10^{-9}
Interpretation:
The Fisher's Exact Test produced a p-value less than 0.05, so we reject the null hypothesis. This indicates a statistically significant association between drug treatment and disease outcome.
The odds ratio is 16.0, meaning the odds of having no disease for Drug A are 16 times higher compared to Drug B.
Conclusion:
Drug A appears to be much more effective than Drug B in preventing disease.

Question Number 5

Problem 5

A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average), and prestige of the undergraduate institution, affect admission into graduate school. The response variable, admit/don't admit, is a binary variable. The dataset is taken from https://stats.idre.ucla.edu/stat/data/binary.csv (stats.idre.ucla.edu in Bing) and fit logistic generalized linear models (GLMs) to identify the effect of admission into graduate school. Interpret the result.

Binary Logistic Regression Model

```
import pandas as pd
import statsmodels.api as sm

# Load dataset
url = "https://stats.idre.ucla.edu/stat/data/binary.csv"
df = pd.read_csv(url)

# Create dummy variables
df = pd.get_dummies(df, columns=["rank"], drop_first=True)

# Define X and y
X = df.drop(columns=["admit"])
y = df["admit"]

# Add constant
X = sm.add_constant(X)

# Logistic regression model
model = sm.Logit(y, X).fit()

# Model summary
print(model.summary())
```

Variable	Coef.	Std. Err.	z	P> $|z|$	[0.025, 0.975]
const	-2.4386	0.896	-2.722	0.006	[-4.194, -0.683]
gre	0.0029	0.001	2.748	0.006	[0.001, 0.005]
gpa	0.2753	0.252	1.093	0.274	[-0.218, 0.769]
rank_2	-0.6902	0.315	-2.192	0.028	[-1.307, -0.073]
rank_3	-1.2829	0.342	-3.754	0.000	[-1.953, -0.613]
rank_4	-1.5653	0.417	-3.756	0.000	[-2.382, -0.749]

Table 19: Logistic Regression Results for Admission Prediction

Interpretation of Logistic Regression Results

- The model is statistically significant overall (LLR p-value ¡ 0.001), meaning at least one predictor affects admission.
- GRE score has a positive and significant effect ($p = 0.006$), so higher GRE increases admission chances.
- GPA is not significant ($p = 0.274$), so it does not strongly affect admission in this model.
- School rank matters strongly:
  - Rank 2 → lower chance than Rank 1 (significant).
  - Rank 3 → much lower chance (highly significant).
  - Rank 4 → lowest chance (highly significant).
- The model explains a small but meaningful variation in admission (Pseudo R^2 0.073).

Variable	p-value
Intercept	1.4688×10^{-31}
Program[T.B]	1.2530×10^{-07}
Program[T.C]	3.0644×10^{-02}

Table 20: Significant Variables from Logistic Regression

Interpretation

- All variables are statistically significant because their p-values are less than 0.05.
- Program B is highly significant ($p = 1.253 \times 10^{-7}$), showing a strong difference from the reference group.
- Program C is also significant ($p = 0.0306$), indicating a meaningful difference from the reference group.
- The intercept is significant, but it mainly represents the baseline level of the model.
- Overall, the program type has a significant effect on the outcome variable.

Question Number 6

Problem 6

The number of awards earned by students at one high school. Predictors of the number of awards earned include the type of program in which the student was enrolled (e.g., vocational, general or academic) and the score on their final exam in math. The data set is taken from https://stats.idre.ucla.edu/stat/data/poisson fit the poisson generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Poisson Regression Model

```
import statsmodels.api as sm
import statsmodels.formula.api as smf

# Poisson regression model
model = smf.glm(
    formula = "num_awards ~ prog + math",
    data = df,
    family = sm.families.Poisson()
).fit()

# Model summary
print(model.summary())
```

Variable	**Coef.**	**Std. Err.**	**z**	**P>** $\lvert z \rvert$	**[0.025, 0.975]**
Intercept	-5.5781	0.677	-8.242	0.000	[-6.905, -4.252]
prog	0.1233	0.163	0.755	0.450	[-0.197, 0.443]
math	0.0861	0.010	8.984	0.000	[0.067, 0.105]

Table 21: Poisson GLM Regression Results for Number of Awards

Interpretation

- The model is statistically significant overall (Pseudo R^2 0.344), explaining a meaningful portion of variation in the number of awards.
- The intercept is highly significant (p ¡ 0.001), representing the baseline log-count of awards.
- The variable **math** has a positive and highly significant effect (p ¡ 0.001). Higher math scores increase the expected number of awards.
- The variable **prog** is not statistically significant ($p = 0.450$), suggesting program type does not strongly affect awards in this model.
- Overall, math achievement is the strongest predictor of awards, while program type shows no meaningful impact.

Variable	**p-value**
Intercept	1.7001×10^{-16}
prog	4.5021×10^{-01}
math	2.6113×10^{-19}

Table 22: P-values of Variables in Poisson GLM

Interpretation

- The **Intercept** is highly significant ($p < 0.001$), representing the baseline log-count of awards.
- The variable **math** is also highly significant ($p < 0.001$), showing that higher math scores strongly increase the expected number of awards.
- The variable **prog** is not statistically significant ($p = 0.450$), meaning program type does not have a meaningful effect in this model.
- Overall, math achievement is the strongest predictor of awards, while program type shows no significant impact.

Question Number 7

Problem 7

Problem 7: School administrators study the attendance behavior of high school juniors at two schools. Predictors of the number of days of absence include the type of program in which the student is enrolled and a standardized test in math. The data set is taken from ("https://stats.idre.ucla.edu/stat/stata/dae/nbdata.dta") and fit the negative binomial generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Negative Binomial Regression Model

```
import statsmodels.api as sm
import statsmodels.formula.api as smf

# Negative Binomial regression
model = smf.glm(
    formula = "daysabs ~ prog + math",
    data = df,
    family = sm.families.NegativeBinomial()
).fit()

# Model summary
print(model.summary())
```

Variable	**Coef.**	**Std. Err.**	**z**	**P>** $\lvert z \rvert$	**[0.025, 0.975]**
Intercept	3.4875	0.227	15.336	0.000	[3.042, 3.933]
prog	-0.6781	0.098	-6.921	0.000	[-0.870, -0.486]
math	-0.0067	0.003	-2.692	0.007	[-0.012, -0.002]

Table 23: Negative Binomial GLM Regression Results for Student Absenteeism

Interpretation

- The Negative Binomial GLM shows that the model is statistically significant and fits the data reasonably well (Pseudo $R^2 = 0.1869$).
- The variable **prog** is highly significant ($p < 0.001$) and has a negative effect, meaning certain programs are associated with fewer absence days.
- The variable **math** is also significant ($p = 0.007$) and negatively related to absences, so higher math scores lead to fewer days absent.
- The **Intercept** is highly significant ($p < 0.001$), representing the baseline expected level of absence when predictors are zero.
- Overall, both program type and math performance significantly reduce student absenteeism.

Variable	**Coefficient**
Intercept	3.4875
prog	-0.6781
math	-0.0067

Table 24: Negative Binomial GLM Coefficients for Student Absenteeism

Interpretation

- The **Intercept** (3.4875) represents the baseline expected log-count of absence days when predictors are zero.
- The variable **prog** has a negative coefficient (-0.6781), meaning certain programs are associated with fewer absence days. This effect is highly significant.
- The variable **math** also has a negative coefficient (-0.0067), indicating that higher math scores reduce absenteeism. This effect is statistically significant.
- Overall, both program type and math performance significantly reduce student absenteeism, while the intercept captures the baseline level.

Question Number 8

Problem 8

The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, were there children in the group and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish so there are excess zeros in the data because of the people that did not fish. The data set is taken from (”https://stats.idre.ucla.edu/stat/data/fish.csv”) and fit the Zero-Inflated Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Listing 1: Poisson Regression Model in Python

```python
import statsmodels.api as sm
import statsmodels.formula.api as smf

model = smf.glm(
    formula="count ~ persons + child + camper",
    data=df,
    family=sm.families.Poisson()
).fit()

print(model.summary())
```

Variable	**Coef.**	**Std. Err.**	**z**	**P> $\|z\|$**	**[0.025, 0.975]**
Intercept	-1.9818	0.152	-13.016	0.000	[-2.280, -1.683]
persons	1.0913	0.039	27.799	0.000	[1.014, 1.168]
child	-1.6900	0.081	-20.866	0.000	[-1.849, -1.531]
camper	0.9309	0.089	10.450	0.000	[0.756, 1.106]

Table 25: Poisson GLM Regression Results for Count Variable.

Interpretation: The model is statistically significant (Pseudo $R^2 = 0.9985$), showing excellent fit.

The variable **persons** has a strong positive effect (p ¡ 0.001), meaning more persons increase the expected count.

The variable **child** has a strong negative effect (p ¡ 0.001), indicating households with children have fewer counts.

The variable **camper** is positive and significant (p ¡ 0.001), showing campers increase the expected count.

The **Intercept** is significant and represents the baseline log-count when predictors are zero.

Variable	p-value
Intercept	9.9479×10^{-39}
persons	4.4431×10^{-170}
child	1.0986×10^{-96}
camper	1.4690×10^{-25}

Table 26: P-values of Variables in Poisson GLM

Interpretation

- All variables are statistically significant because their p-values are far less than 0.05.
- This means **persons**, **child**, and **camper** all have a significant effect on the number of fish caught.
- The extremely small p-value for **persons** (4.44×10^{-170}) shows it is the strongest predictor in the model.
- The **child** variable (1.10×10^{-96}) is also highly significant, indicating it strongly affects fish catch behavior.
- Overall, all predictors meaningfully influence the number of fish caught in the Poisson model.

Variable	Coefficient
Intercept	0.1378
persons	2.9780
child	0.1845
camper	2.5369

Table 27: Coefficients of Variables in Poisson GLM

Interpretation

- The intercept (0.1378) represents the baseline expected log count of fish when all predictors are zero.
- The coefficient of **persons** (2.9780) is strongly positive, meaning larger groups greatly increase fish catch.
- The coefficient of **child** (0.1845) is slightly positive, showing a weak increase in fish catch for groups with children.
- The coefficient of **camper** (2.5369) is also strongly positive, indicating campers catch significantly more fish.
- Overall, group size and camping have a strong positive effect on fish catch, while children have a very small effect.

Question Number 9

Problem 9

The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, were there children in the group and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish so there are excess zeros in the data because of the people that did not fish. The data set is taken from ("https://stats.idre.ucla.edu/stat/data/fish.csv") and fit the Zero-Inflated Binomial regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Listing 2: Binomial GLM Model in Python

```python
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf

url = "https://stats.idre.ucla.edu/stat/data/fish.csv"
df = pd.read_csv(url)

model = smf.glm(
    formula="count ~ persons + child + camper",
    data=df,
    family=sm.families.Binomial()
).fit()

print(model.summary())
```

Variable	**Coef.**	**Std. Err.**	**z**	**P>** $\lvert z\rvert$	**[0.025, 0.975]**
Intercept	-3.1×10^{16}	1.21×10^{07}	-2.56×10^{09}	0.000	$[-3.1\times10^{16}, -3.1\times10^{16}]$
persons	2.01×10^{16}	4.57×10^{06}	4.40×10^{09}	0.000	$[2.01\times10^{16}, 2.01\times10^{16}]$
child	-2.35×10^{16}	5.97×10^{06}	-3.94×10^{09}	0.000	$[-2.35\times10^{16}, -2.35\times10^{16}]$
camper	1.38×10^{16}	8.63×10^{06}	1.59×10^{09}	0.000	$[1.38\times10^{16}, 1.38\times10^{16}]$

Table 28: Binomial GLM Regression Results for Count Variable

Interpretation

- The Binomial GLM shows that all variables are statistically significant ($p < 0.001$).
- The variable **persons** has a very large positive coefficient (2.011e+16), meaning larger groups greatly increase fish catch.
- The variable **child** has a very large negative coefficient (-2.352e+16), meaning groups with children greatly reduce fish catch.
- The variable **camper** also has a large positive coefficient (1.377e+16), indicating campers catch more fish.
- Overall, group size increases fish catch, while children decrease it and camping increases fish catch significantly.

Variable	**Coefficient**
Intercept	-3.1002×10^{16}
persons	2.0113×10^{16}
child	-2.3516×10^{16}
camper	1.3766×10^{16}

Table 29: Binomial GLM Coefficients for Fish Catch Model

Interpretation

- The intercept (-3.10×10^{16}) is extremely negative, representing the baseline log count when predictors are zero.
- The coefficient of **persons** (2.01×10^{16}) is very large and positive, meaning larger groups strongly increase fish catch.
- The coefficient of **child** (-2.35×10^{16}) is very large and negative, indicating children greatly reduce fish catch.
- The coefficient of **camper** (1.38×10^{16}) is also very large and positive, showing campers catch more fish.
- Overall, all predictors have extremely strong effects on fish catch in this model.

Question Number 10

Problem 10

A study of length of hospital stay, in days, as a function of age, kind of health insurance and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from (”https://stats.idre.ucla.edu/stat/data/ztp.dta”) and fit the zero-truncated Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.section*Problem 10

Listing 3: Loading Stata Dataset in Python

```python
import requests
import pyreadstat

url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"
file_path = "ztp.dta"

r = requests.get(url)

with open(file_path, "wb") as f:
    f.write(r.content)

df, meta = pyreadstat.read_dta("ztp.dta")

df.head()
```

Variable	**Coef.**	**Std. Err.**	**z**	**P>** $\lvert z \rvert$	**[0.025, 0.975]**
Intercept	2.4358	0.027	89.149	0.000	[2.382, 2.489]
age	-0.0144	0.005	-2.867	0.004	[-0.024, -0.005]
hmo	-0.1357	0.024	-5.722	0.000	[-0.182, -0.089]
died	-0.2036	0.018	-11.088	0.000	[-0.240, -0.168]

Table 30: Poisson GLM Regression Results for Hospital Stay

Interpretation

- The model is statistically significant overall (Pseudo R^2 = 0.1142), explaining a modest but meaningful portion of variation in hospital stay.
- The **Intercept** (2.4358) is highly significant, representing the baseline expected log-count of hospital stay.
- The variable **age** (-0.0144, p = 0.004) is significant and negative, meaning older patients tend to have slightly shorter stays.
- The variable **hmo** (-0.1357, p ¡ 0.001) is significant and negative, indicating patients with HMO insurance plans have fewer hospital stay days.
- The variable **died** (-0.2036, p ¡ 0.001) is significant and negative, showing patients who died had shorter hospital stays.
- Overall, age, insurance type, and mortality status all significantly reduce hospital stay length in this model.

Variable	**p-value**
Intercept	0.0000
age	0.0041
hmo	1.05×10^{-08}
died	1.43×10^{-28}

Table 31: P-values of Variables in Poisson GLM for Hospital Stay

Interpretation

- All variables are statistically significant because their p-values are far less than 0.05.
- The variable **age** (p = 0.0041) is significant and negative, meaning older patients tend to have slightly shorter hospital stays.
- The variable **hmo** ($p = 1.05 \times 10^{-08}$) is highly significant, showing that patients with HMO insurance plans have fewer hospital stay days.
- The variable **died** ($p = 1.43 \times 10^{-28}$) is extremely significant, indicating patients who died had shorter hospital stays.
- The **Intercept** is significant, representing the baseline expected log-count of hospital stay when predictors are zero.
- Overall, age, insurance type, and mortality status all significantly reduce hospital stay length in this model.

Variable	Coefficient
Intercept	11.4245
age	0.9857
hmo	0.8731
died	0.8158

Table 32: Poisson GLM Coefficients for Hospital Stay

Interpretation

- The **Intercept** (11.42) represents the baseline expected rate of hospital stay.
- The **age** IRR = 0.9857 means that each additional year of age slightly decreases hospital stay by about 1.4%.
- The **hmo** IRR = 0.8731 shows that patients with HMO insurance have about 12.7% lower hospital stay compared to others.
- The **died** IRR = 0.8158 indicates that patients who died had about 18.4% shorter recorded stays.
- Overall, all IRR values below 1 suggest a decreasing effect on hospital stay.

Question Number 11

Problem 11

A study of length of hospital stay, in days, as a function of age, kind of health insurance and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from ("https://stats.idre.ucla.edu/stat/data/ztp.dta") and fit the zero-truncated Negative binomial regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school.

Listing 4: Negative Binomial Regression Model in Python

```
import statsmodels.api as sm
import statsmodels.formula.api as smf

model = smf.glm(
    formula="stay ~ age + hmo + died",
    data=df,
    family=sm.families.NegativeBinomial()
).fit()

print(model.summary())
```

Variable	**Coef.**	**Std. Err.**	**z**	**P>** $\lvert z \rvert$	**[0.025, 0.975]**
Intercept	2.4378	0.091	26.920	0.000	[2.260, 2.615]
age	-0.0147	0.016	-0.894	0.371	[-0.047, 0.018]
hmo	-0.1375	0.075	-1.845	0.065	[-0.284, 0.009]
died	-0.2038	0.058	-3.505	0.000	[-0.318, -0.090]

Table 33: Negative Binomial GLM Regression Results for Hospital Stay

Interpretation

- The **Intercept** (2.4378) is highly significant, representing the baseline expected log-count of hospital stay.
- The variable **age** (-0.0147, p = 0.371) is not statistically significant, meaning age does not have a meaningful effect on hospital stay in this model.
- The variable **hmo** (-0.1375, p = 0.065) is marginally insignificant, suggesting HMO insurance may reduce hospital stay, but the effect is not strongly supported statistically.
- The variable **died** (-0.2038, p ¡ 0.001) is significant and negative, showing patients who died had shorter hospital stays.
- Overall, only mortality status (**died**) significantly reduces hospital stay length, while age and insurance type show weak or insignificant effects.

INTERPRET: The intercept coefficient is 2.437758, representing the baseline log expected hospital stay when all predictors are zero. The coefficient for age is -0.014748, indicating that an increase in age slightly decreases the expected hospital stay. The coefficient for hmo is -0.137485, meaning patients with HMO insurance tend to have shorter hospital stays compared to non-HMO patients. The coefficient for died is -0.203813, showing that patients who died in the hospital had lower expected hospital stay than those who survived. Overall, all predictor coefficients are negative, indicating a decreasing effect on the expected length of hospital stay.

Variable	**p-value**
Intercept	1.27×10^{-159}
age	0.3711
hmo	0.0649
died	4.56×10^{-04}

Table 34: P-values of Variables in Negative Binomial GLM for Hospital Stay

Interpretation

- The **Intercept** ($p < 0.001$) is highly significant, representing the baseline expected hospital stay.
- The variable **age** ($p = 0.3711$) is not statistically significant, meaning age does not have a meaningful effect on hospital stay.
- The variable **hmo** ($p = 0.0649$) is marginally insignificant, suggesting patients with HMO insurance may have slightly shorter hospital stays, but the effect is not significant at the 5% level.
- The variable **died** ($p < 0.001$) is highly significant, showing that patients who died had significantly shorter hospital stays compared to others.
- Overall, the model suggests that only **died** is a statistically significant predictor of hospital stay, while age and insurance type show weak or insignificant effects.